

Math 162
Midterm Exam 1

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1. Let $f : [0, 2] \rightarrow \mathbb{R}$ be the function defined by

$$f(x) := \begin{cases} x^2 & \text{for } x \in \mathbb{Q}; \\ 0 & \text{for } x \notin \mathbb{Q}. \end{cases}$$

- (a) Compute $L(f, P)$ for any partition P of $[0, 2]$.

- (b) Compute $\inf\{U(f, P) \mid P \text{ is a partition of } [0, 2]\}$.

- (c) Explain whether f is integrable using your answers to (a) and (b).

2. If $f : [a, b] \rightarrow \mathbb{R}$ is integrable, prove that

$$\left| \int_a^b f(x) \, dx \right| \leq \int_a^b |f(x)| \, dx.$$

(Hint: what is the relationship between the functions f , $|f|$, and $-|f|$?)

3. Are the following statements true or false? Provide a brief justification for your answer.

(a) Every Darboux sum is also a Riemann sum with respect to some tagging.

(b) If $f : [a, b] \rightarrow \mathbb{R}$ is not integrable, then f is discontinuous at infinitely many points.

(c) There exists a function $f : [0, 1] \rightarrow \mathbb{R}$ such that $f(x) \geq 0$ for all $x \in [0, 1]$, $f(c) > 0$ for some $c \in [0, 1]$, and $\int_0^1 f(x) \, dx = 0$.

(d) If f is a uniformly continuous function on an interval A , then f is bounded.